EXPERIMENT-5

ODD OR EVEN:

AIM:

To write the program for odd or even using R-tool.

PROGRAM:

```
num=as.integer(readline(prompt="enter a number:"))
if((num%%2)==0)
{
 print("number is a even")
}else{
 print("number is odd")
}
```

RESULT:

Thus the basic program odd or even was executed successfully.

```
> num <- as.integer(readline(prompt = "Enter the number: "))
Enter the number: 2
> if(num %% 2 == 0)
+ {print("number is even")}else{print("number is odd")}
[1] "number is even"
> num <- as.integer(readline(prompt = "Enter the number: "))
Enter the number: 5
> if(num %% 2 == 0)
+     + {print("number is even")}else{print("number is odd")}
[1] "number is odd"
```

EXPERIMENT-6

MEAN,MEDIAN,MODE:

AIM:

To write the program for mean,median,mode.

PROGRAM:

MEAN

```
names<-c("siri","mahi","chiru")
age<-c(23,24,25)
marks<-c(88,78,25)
df<-data.frame(names,age,marks)
mean(df $age)
```

MEDIAN

```
names<-c("siri","mahi","chiru")
age<-c(23,24,25)
marks<-c(88,78,25)
df<-data.frame(names,age,marks)
median(df $age)
```

MODE

```
names<-c("siri","mahi","chiru")
age<-c(23,24,25)
marks<-c(88,78,25)
df<-data.frame(names,age,marks)
mode(df $age)
```

RESULT:

Thus the central tendency and measure of dispersion is executed successfully.

EXPERIMENT-7

SUMMARY:

AIM:

To write the program for summary using R-tool.

PROGRAM:

```
names<-c("siri","mahi","chiru")

age<-c(23,24,25)

marks<-c(88,78,25)

df<-data.frame(names,age,marks)

summary(df $age)
```

RESULT:

Thus the central tendancy and measure of dispersion is executed successfully

```
> names=c("siri","mahi","chiru")
> age=c(23,24,25)
> marks=c(88,78,25)
> df=data.frame(names,age,marks)
> mean(df$age)
[1] 24
> median(df$marks)
[1] 78
> mode <- names(sort(table(marks), decreasing = TRUE))[1]
> print(mode)
[1] "25"
> summary(df$age)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   23.0    23.5    24.0    24.0    24.5    25.0
```

EXPERIMENT-8

GREATER AMONG THREE NUMBERS:

AIM:

To write the program for the greatest among three numbers.

PROGRAM:

```
x <- as.integer(readline(prompt = "Enter first number :"))

y <- as.integer(readline(prompt = "Enter second number :"))

z <- as.integer(readline(prompt = "Enter third number :"))


if (x > y && x > z) {

  print(paste("Greatest is :", x))

} else if (y > z) {

  print(paste("Greatest is :", y))

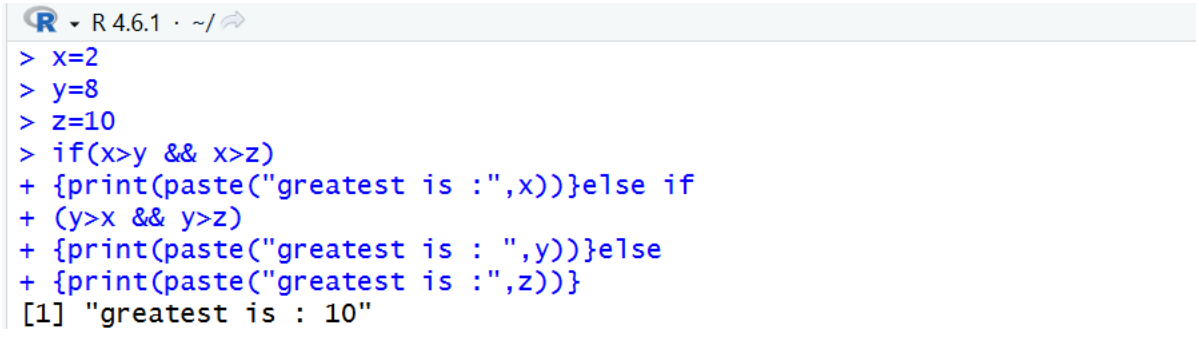
} else{

  print(paste("Greatest is :", z))

}
```

RESULT:

Thus the greatest among the three numbers was executed successfully.



EXPERIMENT-9

IQR:

AIM:

To write the program for central tendency and data dispersion measures using R tool.

PROGRAM:

```
names<-c("siri","mahi","chiru")
age<-c(23,24,25)
marks<-c(88,78,25)
df<-data.frame(names,age,marks)
IQR(df $age)
```

RESULT:

Thus the program for central tendency and data dispersion measures was executed successfully.

EXPERIMENT-10

QUANTILE:

AIM:

To write the program for central tendency and data dispersion measures.

PROGRAM:

```
names<-c("siri","mahi","chiru")
age<-c(23,24,25)
marks<-c(88,78,25)
df<-data.frame(names,age,marks)
quantile(df $age)
```

RESULT:

Thus the program for central tendency and data dispersion measures was executed successfully

EXPERIMENT-11

MID RANGE:

AIM:

To write the program for central tendency and data dispersion measures.

PROGRAM:

names<-c("siri","mahi","chiru")

age<-c(23,24,25)

marks<-c(88,78,25)

df<-data.frame(names,age,marks)

mid range(df $age)

RESULT:

Thus the program for central tendency and data dispersion measures was executed successfully

```
> names=c("siri","mahi","chiru")
> age=c(23,24,25)
> marks=c(88,78,25)
> df=data.frame(names,age,marks)
> mean(df$age)
[1] 24
> median(df$marks)
[1] 78
> mode <- names(sort(table(marks), decreasing = TRUE))[1]
> print(mode)
[1] "25"
> summary(df$age)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   23.0    23.5    24.0    24.0    24.5    25.0
> IQR(df$marks)
[1] 31.5
> quantile(df$age)
  0%  25%  50%  75% 100%
23.0 23.5 24.0 24.5 25.0
> quantile(df$marks)
  0%  25%  50%  75% 100%
25.0 51.5 78.0 83.0 88.0
> mid range(df$age)
Error: unexpected symbol in "mid range"

> midrange=(min(df$marks)+max(df$marks))/2
> print(midrange)
[1] 56.5
> midrange=(min(df$age)+max(df$age))/2
> print(midrange)
[1] 24
```